

# Sharper Theory for Tail-Index Marginal Screening

Mathematical addendum

## 1 What can be weakened, and what can be strengthened

This addendum uses the notation of the manuscript. The main changes are: (i) continuity of the projected conditional distribution is removed by a randomized conditional probability integral transform; (ii) full fibre support is replaced by an exact argmax-accessibility condition; (iii) the screening theorem targets the identifiable set of positive-gap coordinates, without assuming that every active coordinate is marginally visible; (iv) the equicontinuity requirement on the projected-index profiles is replaced by the strictly weaker requirement that the actual quadrature error vanish; and (v) a disjoint-block rank-Hill score is introduced whose stochastic score error is of order  $(n\alpha)^{-1/2}$ , rather than  $(n\alpha h)^{-1/2}$ , while requiring only averaged tail and spatial moduli.

Throughout,  $I_\varepsilon = [\varepsilon, 1 - \varepsilon]$ ,  $0 < \varepsilon < 1/2$ , and  $\lambda$  denotes Lebesgue measure. The dimension, all tuning parameters, and all model objects may form a triangular array.

## 2 Population identification without full fibre support

For a pointwise regular conditional distribution  $K_j(u, dv)$  of  $U_{-j}$  given  $U_j = u$ , put

$$C_j(u) = \text{supp } K_j(u, \cdot) \subseteq [0, 1]^{p-1}.$$

The set  $C_j(u)$  is nonempty and compact. Let  $\mathcal{M} = \arg \max_{x \in [0, 1]^p} \gamma(x)$  and define the *accessible-argmax set*

$$\mathcal{H}_j = \{u \in I_\varepsilon : (\{u\} \times C_j(u)) \cap \mathcal{M} \neq \emptyset\}.$$

The notation  $\{u\} \times C_j(u)$  is understood through the insertion map  $\iota_j$ .

**Theorem 2.1** (Support-restricted envelope and exact marginal target). *Assume the conditional regular-variation and uniform-tail assumptions (C1)–(C2), and assume that  $\gamma$  is continuous. No full-support assumption is imposed. Then, for every pointwise version  $K_j(u, \cdot)$ ,*

$$\xi_j(u) = \text{ess sup}_{v \sim K_j(u, \cdot)} \gamma\{\iota_j(u, v)\} = \max_{v \in C_j(u)} \gamma\{\iota_j(u, v)\}. \quad (1)$$

Consequently,

$$\xi_j(u) = \gamma^* \iff u \in \mathcal{H}_j. \quad (2)$$

If

$$\Psi_j = \frac{1}{\lambda(I_\varepsilon)} \int_{I_\varepsilon} \xi_j(u) du, \quad \Delta_j = \gamma^* - \Psi_j,$$

then

$$\Delta_j > 0 \iff \lambda(I_\varepsilon \setminus \mathcal{H}_j) > 0. \quad (3)$$

No continuity of  $u \mapsto \xi_j(u)$  is needed for (??).

*Proof.* The projection theorem under (C1)–(C2) gives the first expression in (??). Fix  $j, u$  and abbreviate  $K = K_j(u, \cdot)$ ,  $C = C_j(u)$ , and  $f(v) = \gamma\{\iota_j(u, v)\}$ . Since  $C$  is compact and  $f$  is continuous,  $m = \max_C f$  is attained. Because  $K(C) = 1$ ,  $f \leq m$   $K$ -almost surely, so  $\text{ess sup}_K f \leq m$ . Let  $v_\star \in C$  attain  $m$ . For every  $\eta > 0$ , continuity gives an open neighbourhood  $O_\eta$  of  $v_\star$  on which  $f > m - \eta$ . The defining property of the support gives  $K(O_\eta) > 0$ , hence  $\text{ess sup}_K f \geq m - \eta$ . Letting  $\eta \downarrow 0$  proves (??).

The maximum in (??) equals  $\gamma^\star$  exactly when it is attained at a point of the global argmax set, which is (??). Finally,  $0 \leq \gamma^\star - \xi_j(u) \leq \gamma^\star$  and

$$\Delta_j = \frac{1}{\lambda(I_\varepsilon)} \int_{I_\varepsilon} \{\gamma^\star - \xi_j(u)\} du.$$

A nonnegative measurable function has zero integral if and only if it is zero almost everywhere. Combining this fact with (??) gives (??).  $\square$

**Definition 2.2** (Null accessibility and the identifiable set). The model satisfies *null argmax accessibility* when

$$\lambda(I_\varepsilon \setminus \mathcal{H}_j) = 0 \quad \text{for every } j \notin \mathcal{A}. \quad (\text{NA})$$

Define

$$\mathcal{D} = \{j \leq p : \Delta_j > 0\}, \quad \Delta_{\mathcal{D}} = \min_{j \in \mathcal{D}} \Delta_j, \quad (4)$$

with the convention  $\Delta_{\mathcal{D}} = +\infty$  when  $\mathcal{D} = \emptyset$ .

**Corollary 2.3** (Minimal population identification). *Under the assumptions of Theorem ??, condition (NA) is equivalent to  $\mathcal{D} \subseteq \mathcal{A}$ . Moreover  $\mathcal{D} = \mathcal{A}$  if and only if, in addition to (NA),*

$$\lambda(I_\varepsilon \setminus \mathcal{H}_j) > 0 \quad \text{for every } j \in \mathcal{A}.$$

*Full fibre support is sufficient for these conclusions but is not necessary.*

*Proof.* By Theorem ??,  $j \in \mathcal{D}$  is equivalent to  $\lambda(I_\varepsilon \setminus \mathcal{H}_j) > 0$ . Thus (NA) is exactly the statement that no inactive coordinate belongs to  $\mathcal{D}$ . Equality  $\mathcal{D} = \mathcal{A}$  requires and is implied by the additional positive-measure condition for every active coordinate.  $\square$

**Remark 2.4** (Why a dependent inactive proxy can have a genuine marginal signal). *If (NA) fails, an inactive coordinate can restrict the conditional support of the active coordinates in a way that removes near-maximizing configurations on a positive-measure set of its values. Its projected tail index then varies, even though the structural function  $\gamma$  does not depend on that coordinate. This is not an estimation error: the marginal law of  $(U_j, Y)$  genuinely contains a tail-index signal. No coordinatewise marginal method can distinguish such a proxy from a structurally active variable without an accessibility or conditional-independence assumption.*

### 3 Atoms are harmless: randomized conditional PIT

The continuity clause in (E1), and the atomlessness clause in (P1), are not needed. They can be removed without changing the estimator.

**Lemma 3.1** (Randomized conditional probability integral transform). *Let  $F_j(\cdot | u)$  be an arbitrary conditional distribution function of  $Y$  given  $U_j = u$ , possibly with atoms, and let  $Q_j(s, u) = F_j^{-1}(1 - s^{-1} | u)$ . Enlarge the probability space by iid  $Z_{ij} \sim \text{Unif}(0, 1)$ , independent of the data, and define*

$$T_{ij} = F_j(Y_i^- | U_{ij}) + Z_{ij}\{F_j(Y_i | U_{ij}) - F_j(Y_i^- | U_{ij})\}, \quad (5)$$

$$V_{ij} = 1 - T_{ij}. \quad (6)$$

*For every fixed  $j$ , conditionally on  $U_j = (U_{1j}, \dots, U_{nj})$ , the variables  $V_{1j}, \dots, V_{nj}$  are iid standard uniforms and*

$$Y_i = Q_j(V_{ij}^{-1}, U_{ij}) \quad \text{almost surely.} \quad (7)$$

*Consequently every Rényi-spacings argument used by a local Hill proof remains valid without continuity of  $F_j(\cdot | u)$ .*

*Proof.* Fix  $u$  and write  $F = F_j(\cdot | u)$ . For  $t \in (0, 1)$  let  $q = F^{-1}(t)$ . Then  $F(q^-) \leq t \leq F(q)$ . Conditional on  $Y < q$ , the random variable in (??) is at most  $t$ ; conditional on  $Y > q$ , it is larger than  $t$ , apart from null endpoint events. If  $\Delta F(q) = F(q) - F(q^-) > 0$ , then conditional on  $Y = q$  the probability of being at most  $t$  is

$$\frac{t - F(q^-)}{\Delta F(q)}.$$

Hence

$$\mathbb{P}(T_{ij} \leq t | U_{ij} = u) = F(q^-) + \Delta F(q) \frac{t - F(q^-)}{\Delta F(q)} = t.$$

When  $\Delta F(q) = 0$ , necessarily  $F(q^-) = F(q) = t$  and the same identity holds. Thus  $T_{ij}$ , and therefore  $V_{ij}$ , is conditionally uniform. Independence across  $i$  follows from conditional independence of the data and of the auxiliary uniforms.

The distributional transform also satisfies  $F^{-1}(T_{ij} | U_{ij}) = Y_i$  almost surely. Indeed, on an atom at  $Y_i$  we have  $F(Y_i^- | U_{ij}) < T_{ij} < F(Y_i | U_{ij})$  almost surely, and the generalized inverse is exactly  $Y_i$ . At continuity points the only possible failure would place  $Y_i$  in an interval on which the conditional distribution is constant; the union of the relevant open intervals is countable and has conditional probability zero. Since  $T_{ij} = 1 - V_{ij}$ , (??) follows.  $\square$

## 4 Finite-sample screening of the identifiable set

Let  $G = \{r/(n+1) : 1 \leq r \leq n\} \cap I_\varepsilon$ ,  $L_n = |G|$ ,  $\delta_n = (n+1)^{-1}$ , and retain the manuscript's empirical-rank local Hill score  $\widehat{\Psi}_j$ . Put

$$q_n^\circ = \max_{j \leq p} \left| \frac{1}{L_n} \sum_{u \in G} \xi_j(u) - \Psi_j \right|. \quad (8)$$

This is the exact quadrature error. Assuming  $q_n^\circ = o(\Delta_{\mathcal{D}})$  is strictly weaker than imposing a common modulus of continuity on every profile.

For the existing moduli, write

$$b_n(3\alpha) = \max_{j \leq p} \sup_{u \in I_\varepsilon} \sup_{s \geq (3\alpha)^{-1}, t > 1} \frac{|\log\{\ell_j(ts, u)/\ell_j(s, u)\}|}{\log t},$$

$$\omega_n^\uparrow(r, \tau, \alpha) = \max_{j \leq p} \sup_{u \in I_\varepsilon} \sup_{|v-u| \leq r} \sup_{w \in [\tau, 3\alpha]} \left| \log \frac{Q_j(w^{-1}, v)}{Q_j(w^{-1}, u)} \right|.$$

Let  $K_n$  be the deterministic lower bound on the local number of upper order statistics and let  $S_n = 4n + 1$  be the number of rank-window states.

**Theorem 4.1** (Exact finite-sample score and ranking bound). *Assume the quantile representation  $Q_j(s, u) = s^{\xi_j(u)} \ell_j(s, u)$  and use the randomized PIT of Lemma ???. Suppose  $K_n \geq 2$ ,  $0 < \alpha \leq 1/3$ , and  $0 < h \leq 1 - \varepsilon$ . For every  $\eta > 0$ ,  $0 < t \leq 1$ , and  $0 < \tau \leq 3\alpha$ , define*

$$e_n(\eta, t, \tau) = 2\omega_n^\uparrow(h + \eta + \delta_n, \tau, \alpha) + \Gamma t + (1 + t)b_n(3\alpha) + q_n^\circ, \quad (9)$$

$$R_n(\eta, t, \tau) = 2pe^{-2n\eta^2} + pn\tau + 2pS_ne^{-K_nt^2/4} + pS_ne^{-(K_n+1)/4}. \quad (10)$$

Then

$$\mathbb{P} \left\{ \max_{j \leq p} |\hat{\Psi}_j - \Psi_j| > e_n(\eta, t, \tau) \right\} \leq R_n(\eta, t, \tau). \quad (11)$$

On the complementary event, every pair  $j, k$  satisfying

$$\Delta_j - \Delta_k > 2e_n(\eta, t, \tau) \quad (12)$$

obeys  $\hat{\Psi}_j < \hat{\Psi}_k$ . In particular, if  $e_n(\eta, t, \tau) < \Delta_{\mathcal{D}}/2$ , then every coordinate in  $\mathcal{D}$  precedes every coordinate in  $\mathcal{D}^c$ , and therefore

$$\mathbb{P}(\mathcal{D} \subseteq \hat{\mathcal{A}}_d) \geq 1 - R_n(\eta, t, \tau), \quad d \geq |\mathcal{D}|, \quad (13)$$

and

$$\mathbb{P}(\hat{\mathcal{A}}_{|\mathcal{D}|} = \mathcal{D}) \geq 1 - R_n(\eta, t, \tau). \quad (14)$$

Under (NA),  $\mathcal{D} \subseteq \mathcal{A}$ ; if also  $\mathcal{D} = \mathcal{A}$ , these are the usual sure screening and exact-recovery conclusions.

*Proof.* Condition on a covariate column  $\mathbf{U}_j$  and use Lemma ???. The rank windows are then deterministic and the randomized PIT variables are iid uniforms. The sorting, tail-freezing, Rényi representation, and slowly varying remainder arguments are therefore unchanged. A union bound over coordinates and the at most  $S_n$  admissible window states gives the profile bound

$$\mathbb{P} \left\{ \max_j \sup_{u \in I_\varepsilon} |\hat{\xi}_j(u) - \xi_j(u)| > 2\omega_n^\uparrow(h + \eta + \delta_n, \tau, \alpha) + \Gamma t + (1 + t)b_n(3\alpha) \right\} \leq R_n(\eta, t, \tau).$$

Pathwise,

$$|\hat{\Psi}_j - \Psi_j| \leq \sup_{u \in I_\varepsilon} |\hat{\xi}_j(u) - \xi_j(u)| + q_n^\circ,$$

which proves (??).

Since  $\Psi_j = \gamma^* - \Delta_j$ , on the event in (??),

$$\hat{\Psi}_j - \hat{\Psi}_k \leq -(\Delta_j - \Delta_k) + 2e_n(\eta, t, \tau).$$

This is negative under (??). For  $j \in \mathcal{D}$  and  $k \in \mathcal{D}^c$ ,  $\Delta_j \geq \Delta_{\mathcal{D}}$  and  $\Delta_k = 0$ , yielding strict separation when  $e_n < \Delta_{\mathcal{D}}/2$ . The screening and recovery statements are then deterministic consequences of ranking.  $\square$

**Corollary 4.2** (Sharper asymptotic conditions for the implemented score). *Assume the hypotheses of Theorem ?? and, with  $\tau_n = (pn)^{-2}$ ,*

$$\log(2p) = o(nh^2), \quad (15)$$

$$\log(pn) = o(n\alpha h), \quad (16)$$

$$b_n(3\alpha) + \omega_n^\dagger(C_0h, \tau_n, \alpha) + q_n^\circ = o(\Delta_{\mathcal{D}}) \quad (17)$$

for some  $C_0 > 1$ , and

$$\log(pn) = o(n\alpha h \Delta_{\mathcal{D}}^2). \quad (18)$$

Then every coordinate in  $\mathcal{D}$  precedes every coordinate in  $\mathcal{D}^c$  with probability tending to one, so

$$\mathbb{P}(\mathcal{D} \subseteq \hat{\mathcal{A}}_d) \rightarrow 1 \quad (d \geq |\mathcal{D}|), \quad \mathbb{P}(\hat{\mathcal{A}}_{|\mathcal{D}|} = \mathcal{D}) \rightarrow 1.$$

The rank-displacement condition is (??), involving  $\log p$  rather than  $\log(pn)$ .

*Proof.* Condition (??) permits a sequence  $r_n \rightarrow \infty$  such that  $\log(2p) + r_n = o(nh^2)$ . Set

$$\eta_n = \sqrt{\frac{\log(2p) + r_n}{2n}}.$$

Then  $2pe^{-2n\eta_n^2} = e^{-r_n}$  and  $\eta_n + \delta_n = o(h)$ ; the latter also uses  $nh \rightarrow \infty$ , which follows from (??). Hence  $h + \eta_n + \delta_n \leq C_0h$  eventually.

Similarly choose  $s_n \rightarrow \infty$  with  $\log(2pS_n) + s_n = o(K_n)$  and put

$$t_n = 2\sqrt{\frac{\log(2pS_n) + s_n}{K_n}}.$$

Then the spacings term in (??) tends to zero,  $t_n = o(1)$ , and

$$t_n = O\left(\sqrt{\frac{\log(pn)}{n\alpha h}}\right).$$

The threshold term also vanishes because  $\log(pS_n) = o(K_n)$ , and  $pn\tau_n = (pn)^{-1} \rightarrow 0$ . By (??), the stochastic term is  $o(\Delta_{\mathcal{D}})$ ; the remaining terms are  $o(\Delta_{\mathcal{D}})$  by (??). Theorem ?? applies.  $\square$

**Proposition 4.3** (Bounded variation is enough for quadrature). *Suppose  $0 \leq \xi_j \leq \Gamma$  and  $\max_{j \leq p} \text{TV}_{I_\varepsilon}(\xi_j) \leq V_n$ . Then the full-grid quadrature error in (??) satisfies*

$$q_n^\circ \leq C_\varepsilon \frac{V_n + \Gamma}{n}. \quad (19)$$

Thus no continuity or equicontinuity of  $u \mapsto \xi_j(u)$  is required; a uniformly bounded number and size of jumps is allowed.

*Proof.* Associate to each grid point in  $G$  its Voronoi cell in  $I_\varepsilon$ . Every cell has length  $\delta_n$  except for two boundary perturbations of order  $\delta_n$ . On a cell  $C$  with selected point  $u_C$ ,

$$\int_C |\xi_j(u) - \xi_j(u_C)| du \leq \lambda(C) \text{TV}_C(\xi_j).$$

Summing over disjoint cells gives at most  $\delta_n V_n$ . The discrepancy between equal grid weights and normalized cell lengths is supported by the two boundary cells and is  $O_\varepsilon(\delta_n)$ ; boundedness by  $\Gamma$  therefore contributes  $O_\varepsilon(\Gamma\delta_n)$ . This proves (??).  $\square$

## 5 Exact recovery without knowing the active-set size

The maximum population score is the null baseline whenever at least one zero-gap coordinate exists. This gives a thresholded selector that does not need  $s$  or  $|\mathcal{D}|$ .

**Theorem 5.1** (Baseline thresholding). *Assume  $\mathcal{D}^c \neq \emptyset$  and define*

$$\hat{\gamma}_{\max} = \max_{1 \leq j \leq p} \hat{\Psi}_j, \quad \hat{\Delta}_j = \hat{\gamma}_{\max} - \hat{\Psi}_j.$$

*On any event on which  $\max_j |\hat{\Psi}_j - \Psi_j| \leq e$ , one has*

$$\hat{\Delta}_j \leq 2e, \quad j \in \mathcal{D}^c, \quad (20)$$

$$\hat{\Delta}_j \geq \Delta_j - 2e, \quad j \in \mathcal{D}. \quad (21)$$

*Consequently, for every threshold  $\lambda$  satisfying*

$$2e < \lambda < \Delta_{\mathcal{D}} - 2e, \quad (22)$$

$$\{j : \hat{\Delta}_j > \lambda\} = \mathcal{D}.$$

*Combining this with Theorem ?? yields an explicit finite-sample recovery probability without assuming that  $|\mathcal{D}|$  is known.*

*Proof.* All population scores are at most  $\gamma^*$ , and a coordinate in  $\mathcal{D}^c$  has score exactly  $\gamma^*$ . Hence on the stated event

$$\gamma^* - e \leq \hat{\gamma}_{\max} \leq \gamma^* + e.$$

For  $j \in \mathcal{D}^c$ ,  $\hat{\Psi}_j \geq \gamma^* - e$ , giving (??). For  $j \in \mathcal{D}$ ,  $\hat{\Psi}_j \leq \gamma^* - \Delta_j + e$ , giving (??). The threshold conclusion follows immediately.  $\square$

## 6 A disjoint-block score with an $(n\alpha)^{-1/2}$ stochastic rate

The implemented score averages  $O(n)$  highly overlapping local Hill statistics and controls that average by the maximum profile error. This loses the averaging gain. The following rank-block construction uses disjoint local samples, so its score fluctuation is governed by the total number  $n\alpha$  of extremes used across the trimmed interval.

Write the rank indices in  $G$  as  $r_1 < \dots < r_{L_n}$ . Choose an integer  $m = m_n$  with  $2 \leq m \leq L_n$  and put

$$B = B_n = \lfloor L_n/m \rfloor, \quad k = k_n = \lfloor \alpha m \rfloor \geq 2.$$

For  $b = 1, \dots, B$ , set

$$\begin{aligned} \mathcal{R}_b &= \{r_{(b-1)m+1}, \dots, r_{bm}\}, \\ u_b &= \frac{r_{(b-1)m+1} + r_{bm}}{2(n+1)}, \\ \mathcal{B}_{jb} &= \{i : R_{ij} \in \mathcal{R}_b\}. \end{aligned}$$

The sets  $\mathcal{B}_{j1}, \dots, \mathcal{B}_{jB}$  are disjoint and each has exactly  $m$  observations. Let  $H_{jb}$  be the ordinary Hill statistic based on the largest  $k$  responses in  $\mathcal{B}_{jb}$ , and define

$$\tilde{\Psi}_j = \frac{1}{B} \sum_{b=1}^B H_{jb}. \quad (23)$$

Put

$$\bar{h}_m = \frac{m-1}{2(n+1)}, \quad q_{n,B} = \max_{j \leq p} \left| \frac{1}{B} \sum_{b=1}^B \xi_j(u_b) - \Psi_j \right|. \quad (24)$$

For each block define the local tail and spatial moduli

$$b_{j,b} = \sup_{s \geq (3\alpha)^{-1}, t > 1} \frac{|\log\{\ell_j(ts, u_b)/\ell_j(s, u_b)\}|}{\log t}, \quad (25)$$

$$\omega_{j,b}(r, \tau, \alpha) = \sup_{\substack{v \in [0,1] \\ |v-u_b| \leq r}} \sup_{w \in [\tau, 3\alpha]} \left| \log \frac{Q_j(w^{-1}, v)}{Q_j(w^{-1}, u_b)} \right|, \quad (26)$$

and their averaged versions

$$\bar{b}_n = \max_{j \leq p} \frac{1}{B} \sum_{b=1}^B b_{j,b}, \quad \bar{\omega}_n(r, \tau, \alpha) = \max_{j \leq p} \frac{1}{B} \sum_{b=1}^B \omega_{j,b}(r, \tau, \alpha). \quad (27)$$

These assumptions permit a small collection of rough locations, provided their average contribution is negligible.

**Lemma 6.1** (Weighted exponential-means concentration). *Let  $L_1, \dots, L_B$  be independent, each distributed as the mean of  $k$  iid standard exponentials, and let  $a_b \geq 0$ . Put*

$$Z = \sum_{b=1}^B a_b(L_b - 1), \quad V = \frac{1}{k} \sum_{b=1}^B a_b^2, \quad c = \frac{1}{k} \max_b a_b.$$

Then, for every  $x > 0$ ,

$$\mathbb{P}\{Z > \sqrt{2Vx} + cx\} \leq e^{-x}, \quad \mathbb{P}\{Z < -\sqrt{2Vx}\} \leq e^{-x}. \quad (28)$$

In particular, if  $a_b \leq \Gamma/B$ , then

$$\mathbb{P}\left\{|Z| > \Gamma \left( \sqrt{\frac{2x}{Bk}} + \frac{x}{Bk} \right)\right\} \leq 2e^{-x}. \quad (29)$$

*Proof.* For  $0 < \theta < c^{-1}$ , with  $z_b = \theta a_b/k$ ,

$$\begin{aligned} \log \mathbb{E} e^{\theta Z} &= \sum_{b=1}^B k \{-z_b - \log(1 - z_b)\} \\ &\leq \sum_{b=1}^B \frac{k z_b^2}{2(1 - z_b)} \leq \frac{\theta^2 V}{2(1 - c\theta)}. \end{aligned}$$

The first inequality follows from  $-z - \log(1 - z) \leq z^2/[2(1 - z)]$  for  $0 \leq z < 1$ . The standard Chernoff optimization for this sub-gamma bound gives the upper tail in (??). For the lower tail,

$$\log \mathbb{E} e^{-\theta Z} = \sum_b k \{z_b - \log(1 + z_b)\} \leq \frac{\theta^2 V}{2},$$

which gives the stated sub-Gaussian lower tail. If  $a_b \leq \Gamma/B$ , then  $V \leq \Gamma^2/(Bk)$  and  $c \leq \Gamma/(Bk)$ , proving (??).  $\square$

**Theorem 6.2** (Finite-sample block-score concentration). *Assume the quantile representation  $Q_j(s, u) = s^{\xi_j(u)} \ell_j(s, u)$  and use the randomized PIT of Lemma ???. Suppose  $0 < \alpha \leq 1/3$  and  $k \geq 2$ . For every  $\eta > 0$ ,  $x > 0$ , and  $0 < \tau \leq 3\alpha$ , put*

$$r_{n,B}(\eta, x, \tau) = 2\bar{\omega}_n(\bar{h}_m + \eta + \delta_n, \tau, \alpha) + 2\bar{b}_n + q_{n,B} + \Gamma \left( \sqrt{\frac{2x}{Bk}} + \frac{x}{Bk} \right). \quad (30)$$

Then

$$\begin{aligned} \mathbb{P} \left\{ \max_{j \leq p} |\tilde{\Psi}_j - \Psi_j| > r_{n,B}(\eta, x, \tau) \right\} \\ \leq 2pe^{-2n\eta^2} + pn\tau + 3pBe^{-k/4} + 2pe^{-x}. \end{aligned} \quad (31)$$

Thus the stochastic part of the score error depends on  $Bk$ , the total number of upper-order-statistic contributions across the disjoint blocks.

*Proof.* Fix a coordinate  $j$  and condition on  $\mathbf{U}_j$ . The rank blocks are then deterministic and disjoint. By Lemma ??, their PIT collections are independent and standard uniform. Let  $V_{jb,(1)} < \dots < V_{jb,(m)}$  be the PIT order statistics in block  $b$  and put

$$L_{jb} = \frac{1}{k} \sum_{r=1}^k \log \frac{V_{jb,(k+1)}}{V_{jb,(r)}}.$$

The variables  $L_{j1}, \dots, L_{jB}$  are conditionally independent and each is a mean of  $k$  iid standard exponentials.

Let  $D_n(\eta)$  be the simultaneous rank-displacement event. Its complement has probability at most  $2pe^{-2n\eta^2}$ . On  $D_n(\eta)$ , every observation in block  $b$  satisfies

$$|U_{ij} - u_b| \leq \bar{h}_m + \eta + \delta_n.$$

Outside the event that some randomized PIT is below  $\tau$ , whose probability is at most  $pn\tau$ , and outside the event  $V_{jb,(k+1)} > 3\alpha$ , tail-only freezing gives

$$|H_{jb} - \check{H}_{jb}| \leq 2\omega_{j,b}(\bar{h}_m + \eta + \delta_n, \tau, \alpha),$$

where  $\check{H}_{jb}$  freezes all block quantiles at  $u_b$ . The threshold order-statistic inequality and a union bound give total failure probability at most  $pBe^{-(k+1)/4} \leq pBe^{-k/4}$ .

For the frozen statistic,

$$\check{H}_{jb} = \xi_j(u_b)L_{jb} + R_{jb}, \quad |R_{jb}| \leq b_{j,b}L_{jb}.$$

The exponential Chernoff bound gives  $\mathbb{P}(L_{jb} > 2 \mid \mathbf{U}_j) \leq 2e^{-k/4}$ . Hence, outside an additional event of probability at most  $2pBe^{-k/4}$ ,

$$\frac{1}{B} \sum_{b=1}^B |R_{jb}| \leq \frac{2}{B} \sum_{b=1}^B b_{j,b} \leq 2\bar{b}_n$$

for every  $j$ .



It remains to control

$$Z_j = \frac{1}{B} \sum_{b=1}^B \xi_j(u_b)(L_{jb} - 1).$$

Apply Lemma ?? conditionally with  $a_b = \xi_j(u_b)/B \leq \Gamma/B$ , then take a union bound over  $j$ . With conditional, and hence unconditional, probability at least  $1 - 2pe^{-x}$ ,

$$\max_{j \leq p} |Z_j| \leq \Gamma \left( \sqrt{\frac{2x}{Bk}} + \frac{x}{Bk} \right).$$

Combining the spatial, tail, stochastic, and quadrature terms proves (??).  $\square$

**Corollary 6.3** (Block-score rate and screening). *Suppose*

$$m_n \asymp nh_n, \quad B_n \asymp h_n^{-1}, \quad k_n \asymp n\alpha_n h_n,$$

so that  $B_n k_n \asymp n\alpha_n$ . Assume

$$\log(2p) = o(nh_n^2), \tag{32}$$

$$\log(p/h_n) = o(n\alpha_n h_n), \tag{33}$$

$$\bar{\omega}_n(Ch_n, \tau_n, \alpha_n) + \bar{b}_n + q_{n,B} = o(\Delta_{\mathcal{D}}), \quad \tau_n = (pn)^{-2}, \tag{34}$$

$$\log(2p) = o(n\alpha_n \Delta_{\mathcal{D}}^2). \tag{35}$$

Then

$$\max_{j \leq p} |\tilde{\Psi}_j - \Psi_j| = O_{\mathbb{P}} \left( \sqrt{\frac{\log(2p)}{n\alpha_n}} + \frac{\log(2p)}{n\alpha_n} \right) + o(\Delta_{\mathcal{D}}), \tag{36}$$

and every coordinate in  $\mathcal{D}$  precedes every coordinate in  $\mathcal{D}^c$  with probability tending to one. Hence the sure-screening, exact-recovery, and baseline-threshold conclusions of the previous section hold with  $\tilde{\Psi}$  in place of  $\hat{\Psi}$ .

*Proof.* Choose  $\eta_n$  exactly as in Corollary ??; (??) makes the rank term vanish and ensures that the spatial radius is at most  $Ch_n$ . Since  $B_n \asymp h_n^{-1}$ , (??) gives  $pB_n e^{-k_n/4} \rightarrow 0$ . The PIT truncation term vanishes for  $\tau_n = (pn)^{-2}$ . Finally choose  $x_n = \log(2p) + r_n$ , where  $r_n \rightarrow \infty$  sufficiently slowly. The last term in (??) vanishes, and because  $B_n k_n \asymp n\alpha_n$ , the stochastic radius has the order displayed in (??). Conditions (??)–(??) make the total error  $o_{\mathbb{P}}(\Delta_{\mathcal{D}})$ , after which deterministic separation applies.  $\square$

**Proposition 6.4** (Block quadrature under bounded variation). *If  $0 \leq \xi_j \leq \Gamma$  and  $\max_j \text{TV}_{I_\varepsilon}(\xi_j) \leq V_n$ , then*

$$q_{n,B} \leq C_\varepsilon(V_n + \Gamma) \left( \frac{m_n}{n} + \frac{1}{n} \right). \tag{37}$$

*In the usual choice  $m_n \asymp nh_n$ , the quadrature error is  $O\{(V_n + \Gamma)h_n\}$ , even for discontinuous profiles.*

*Proof.* Let  $d_n = m_n/(n+1)$  and associate to every midpoint  $u_b$  the interval of length  $d_n$  centred at  $u_b$ . These intervals are disjoint and contiguous. Their union differs from  $I_\varepsilon$  only by the two grid-endpoint errors and by the unused final group of fewer than  $m_n$  ranks; its symmetric-difference length is  $O(d_n + \delta_n)$ . On each block interval  $J_b$ ,

$$\int_{J_b} |\xi_j(u) - \xi_j(u_b)| du \leq d_n \text{TV}_{J_b}(\xi_j).$$

Summing gives at most  $d_n V_n$ . Normalizing by an interval length bounded away from zero, and controlling the omitted boundary set by  $\Gamma$ , yields (??).  $\square$

## 7 The primitive doubling condition can be replaced

The manuscript's deficit-mass doubling assumption is a convenient route to a  $1/t$  bound, but it is stronger than the transfer proof needs. Define, with  $D_{j,u} = 1/\gamma - 1/\xi_j(u)$ ,

$$\begin{aligned} B_{j,u}(t) &= \int e^{-tD_{j,u}(v)} c\{\iota_j(u, v)\} K_j(u, dv), \\ M_{j,u}(t) &= \int D_{j,u}(v) e^{-tD_{j,u}(v)} c\{\iota_j(u, v)\} K_j(u, dv), \\ \chi_n(T) &= \max_{j \leq p_n} \sup_{u \in I_{\varepsilon/2}} \sup_{t \geq T} \frac{M_{j,u}(t)}{B_{j,u}(t)}. \end{aligned}$$

**(P3-L)** *Uniform Laplace localization:*  $\lim_{T \rightarrow \infty} \sup_n \chi_n(T) = 0$ .

This condition says exactly that the exponentially tilted fibre law concentrates near zero reciprocal-index deficit. It imposes no doubling, regular variation, density, or polynomial margin.

**Theorem 7.1** (Primitive transfer under uniform Laplace localization). *Assume the geometry and Lipschitz clauses of (P1), but not atomlessness; assume (P2), (P4), and (P3-L). Let  $x_\alpha = \log\{(3\alpha_n)^{-1}\}$  and  $c_\gamma = \underline{\gamma}/2$ . Then, after a common onset,*

$$b_n(3\alpha_n) \leq 6\Gamma^2 \chi_n(c_\gamma x_\alpha) + 4\Gamma^2 \bar{R}_1(c_\gamma x_\alpha). \quad (38)$$

*The spatial transfer bound obtained from (P4) is unchanged, and the conditional-quantile representation follows from Lemma ??; no atomlessness assumption is required. Consequently, the score and screening theorems apply whenever*

$$\chi_n(c_\gamma x_\alpha) + \bar{R}_1(c_\gamma x_\alpha) = o(\Delta_{\mathcal{D}}), \quad (39)$$

*together with the corresponding spatial and quadrature conditions. The original doubling assumption implies (P3-L) with  $\chi_n(T) = O(T^{-1})$ , but is not necessary.*

*Proof.* Suppress  $(j, u)$ . Write

$$\mathcal{A}(t) = \int e^{-tD(v)} c(v) \{1 + r(t, v)\} K(dv).$$

Then the projected survival function factors exactly as

$$\bar{F}_j(e^t \mid u) = e^{-t/\xi_j(u)} \mathcal{A}(t).$$

For large  $t$ ,  $R_0(t) \leq 1/2$ , and differentiation gives

$$\mathcal{A}'(t) = - \int D e^{-tD} c(1 + r) dK + \int e^{-tD} c \partial_t r dK.$$

Since  $(1 - R_0)B \leq \mathcal{A} \leq (1 + R_0)B$ ,

$$\left| \frac{\mathcal{A}'(t)}{\mathcal{A}(t)} \right| \leq 3 \frac{M(t)}{B(t)} + 2R_1(t) \leq 3\chi_n(t) + 2R_1(t) =: g_n(t). \quad (40)$$

In particular  $g_n(t) \rightarrow 0$  uniformly.

It remains to justify a uniform lower linear bound for the log-quantile. Because  $B'(t) = -M(t)$ ,

$$-\{\log B(t)\}' = \frac{M(t)}{B(t)}.$$

Given  $\eta > 0$ , (P3-L) gives a common  $T_\eta$  such that this derivative is at most  $\eta$  for  $t \geq T_\eta$ . Also  $B(T_\eta) \geq c_- e^{-T_\eta/\underline{\gamma}}$  uniformly. Therefore

$$-\log B(t) \leq C_\eta + \eta t, \quad t \geq T_\eta,$$

and hence  $|\log \mathcal{A}(t)| = o(t)$  uniformly. If  $T_{j,u}(x) = \log Q_j(e^x, u)$ , quantile inversion gives

$$x = \frac{T_{j,u}(x)}{\xi_j(u)} - \log \mathcal{A}\{T_{j,u}(x)\}.$$

For all sufficiently large  $x$ , the  $o(T)$  term is bounded in absolute value by  $T/\underline{\gamma}$ , so  $T_{j,u}(x) \geq c_\gamma x$  uniformly.

Let  $a(t) = \mathcal{A}'(t)/\mathcal{A}(t)$ . Once  $|\Gamma|a(t)| \leq 1/2$ , differentiation of the inverse relation yields

$$\frac{d}{dx} \log \ell_j(e^x, u) = \frac{\xi_j(u)^2 a\{T_{j,u}(x)\}}{1 - \xi_j(u) a\{T_{j,u}(x)\}}.$$

Using (??) and  $T_{j,u}(x) \geq c_\gamma x$  gives

$$\sup_{j,u} \left| \frac{d}{dx} \log \ell_j(e^x, u) \right| \leq 6\Gamma^2 \chi_n(c_\gamma x) + 4\Gamma^2 \bar{R}_1(c_\gamma x).$$

The normalized increment defining  $b_n(3\alpha_n)$  is an average of this derivative over an interval starting at  $x \geq x_\alpha$ , proving (??). The spatial argument uses only (P4) and the inverse slope bound, so it is unchanged. Finally, the original doubling condition gives  $M(t)/B(t) = O(1/t)$  by the tilted-deficit estimate, hence implies (P3-L).  $\square$

**Proposition 7.2** (A still weaker near-max mass criterion). *Let*

$$H_{j,u}(z) = \int \mathbf{1}_{\{D_{j,u}(v) \leq z\}} c\{\iota_j(u, v)\} K_j(u, dv), \quad h_n(z) = \inf_{j,u} H_{j,u}(z).$$

For every  $z > 0$  and  $T > 0$ ,

$$\chi_n(T) \leq z + \frac{D_{\max} c_+ e^{-Tz/2}}{h_n(z/2)}, \quad D_{\max} = \underline{\gamma}^{-1}. \quad (41)$$

In particular, if for every fixed  $z > 0$ ,  $\inf_n h_n(z) > 0$ , then (P3-L) holds. This requires only a uniformly positive amount of fibre mass in every fixed neighbourhood of the maximal tail index; it imposes no local shape as  $z \downarrow 0$ .

*Proof.* Again suppress  $(j, u)$ . Split the tilted first moment at  $z$ :

$$M(T) \leq zB(T) + D_{\max} c_+ e^{-Tz}.$$

On the other hand,

$$B(T) \geq e^{-Tz/2} H(z/2).$$

Dividing proves (??). First choose  $z$  small and then let  $T \rightarrow \infty$  to obtain uniform Laplace localization under the stated fixed neighbourhood-mass condition.  $\square$

## 8 An information-theoretic $1/\log n$ lower bound under (C1)–(C2)

The logarithmic scale in the manuscript is not merely an artefact of the Hill proof when the primitive class has no uniform tail onset. The next result uses the full data, so it is stronger than a lower bound for marginal procedures.

**Theorem 8.1** (Delayed-onset non-identifiability). *Fix  $\gamma_0 > 0$  and  $a > 0$ . There are two triangular arrays of models  $P_{0,n}$  and  $P_{1,n}$  with  $p = 2$  which both satisfy (C1)–(C2), full fibre support, atomlessness, and uniform bounds  $0 < \underline{\gamma} \leq \gamma \leq \Gamma$  and  $0 < c_- \leq c(u) \leq c_+ < \infty$ , such that:*

1. *coordinate 1 is inactive under  $P_{0,n}$  and detectable under  $P_{1,n}$ ;*
2. *under  $P_{1,n}$  its population gap satisfies*

$$\Delta_{1,n} = \frac{a}{4\gamma_0 \log n}; \quad (42)$$

3. *the total variation distance between the full  $n$ -sample experiments satisfies*

$$\|P_{0,n}^{\otimes n} - P_{1,n}^{\otimes n}\|_{\text{TV}} \leq n^{-1}. \quad (43)$$

Consequently, for every screening rule  $\hat{\mathcal{A}}_n$  based on the complete sample  $(U_i, Y_i)_{i=1}^n$ ,

$$P_{0,n}(1 \in \hat{\mathcal{A}}_n) + P_{1,n}(1 \notin \hat{\mathcal{A}}_n) \geq 1 - n^{-1}. \quad (44)$$

Thus no procedure can uniformly distinguish an inactive coordinate from a detectable coordinate with gap of order  $1/\log n$  over the unquantified (C1)–(C2) class.

*Proof.* Let  $U_1, U_2$  be independent standard uniforms under both models and set

$$T_n = n^{2\gamma_0}, \quad \delta_n = \frac{a}{\log T_n} = \frac{a}{2\gamma_0 \log n}.$$

Under  $P_{0,n}$ , let  $\gamma_{0,n}(u_1, u_2) = \gamma_0$  and let

$$\bar{F}_{0,n}(y | u) = y^{-1/\gamma_0}, \quad y \geq 1.$$

Under  $P_{1,n}$ , let

$$\gamma_{1,n}(u_1, u_2) = \gamma_0 + \delta_n u_1$$

and define the conditional survival function by the delayed splice

$$\bar{F}_{1,n}(y | u) = \begin{cases} y^{-1/\gamma_0}, & 1 \leq y \leq T_n, \\ T_n^{-1/\gamma_0} (y/T_n)^{-1/\gamma_{1,n}(u)}, & y > T_n. \end{cases} \quad (45)$$

This is continuous and decreasing. For  $y > T_n$  it can be written as

$$\bar{F}_{1,n}(y | u) = c_n(u) y^{-1/\gamma_{1,n}(u)}, \quad c_n(u) = T_n^{1/\gamma_{1,n}(u) - 1/\gamma_0}.$$

Moreover  $c_n(u) \leq 1$  and

$$\log c_n(u) = -\frac{\delta_n u_1 \log T_n}{\gamma_0 \{\gamma_0 + \delta_n u_1\}} \geq -\frac{a}{\gamma_0^2},$$

so  $e^{-a/\gamma_0^2} \leq c_n(u) \leq 1$  uniformly. The slowly varying factor is exactly equal to  $c_n(u)$  beyond  $T_n$ , hence (C1)–(C2) hold. For large  $n$ , the tail indices also obey any fixed positive lower and upper bounds containing  $\gamma_0$ .

Under  $P_{0,n}$  coordinate 1 is inactive. Under  $P_{1,n}$  it is active and, by full support,

$$\xi_{1,n}(u) = \gamma_0 + \delta_n u, \quad \gamma_n^* = \gamma_0 + \delta_n.$$

Since the average of  $u$  on  $I_\varepsilon = [\varepsilon, 1 - \varepsilon]$  is  $1/2$ ,

$$\Delta_{1,n} = \delta_n \left(1 - \frac{1}{2}\right) = \frac{a}{4\gamma_0 \log n},$$

which is (??).

The two one-observation laws agree exactly on the event  $\{Y \leq T_n\}$ , and under both models

$$\mathbb{P}(Y > T_n \mid U) = T_n^{-1/\gamma_0} = n^{-2}.$$

Therefore the two product measures agree on  $E_n = \{\max_{1 \leq i \leq n} Y_i \leq T_n\}$ , which has the same probability under both, and

$$\|P_{0,n}^{\otimes n} - P_{1,n}^{\otimes n}\|_{\text{TV}} \leq P_{0,n}(E_n^c) \leq n n^{-2} = n^{-1}.$$

This proves (??). Applying the elementary testing inequality  $P_0(A) + P_1(A^c) \geq 1 - \|P_0 - P_1\|_{\text{TV}}$  to  $A = \{1 \in \hat{\mathcal{A}}_n\}$  gives (??).  $\square$

**Remark 8.2** (What the lower bound does and does not settle). *Theorem ?? shows that the order  $1/\log n$  is information-theoretically inaccessible uniformly over (C1)–(C2), because those conditions permit the asymptotic tail regime to start after the sample has effectively ended. It does not rule out detecting such gaps under a common quantitative onset condition such as (P2), or after an explicit second-order bias correction. It identifies precisely why a quantified primitive tail condition is indispensable for any uniform theorem.*

## 9 A sharp impossibility result for all coordinatewise marginal screens

Flatness of the projected tail index is not merely a weakness of the average score. In general, even the entire marginal law of  $(U_j, Y)$  can be identical under a model where  $j$  is inactive and a model where it is active.

**Theorem 9.1** (Exact marginal non-identifiability). *There exist two models satisfying (C1)–(C2), full fibre support, continuity of  $\gamma$ , and exact conditional Pareto tails, such that coordinate 1 is inactive in the first model and active in the second, while the law of  $(U_1, Y)$  is exactly the same in both models. Consequently, for every possibly randomized decision rule  $\phi_n = \phi_n\{(U_{i1}, Y_i)_{i=1}^n\} \in \{0, 1\}$ ,*

$$\mathbb{P}_0(\phi_n = 1) + \mathbb{P}_1(\phi_n = 0) = 1, \tag{46}$$

*so its worst-case error over the two models is at least  $1/2$  for every  $n$ .*

*Proof.* Let  $U_1, U_2$  be independent standard uniforms and choose

$$g(t) = \gamma_0 + a \cos(2\pi t), \quad 0 < a < \gamma_0.$$

Under model  $P_0$ , set  $\gamma_0(u_1, u_2) = g(u_2)$ . Under model  $P_1$ , set

$$\gamma_1(u_1, u_2) = g\{(u_1 + u_2) \bmod 1\}.$$

Because  $g$  is periodic,  $\gamma_1$  is continuous on the square. Generate  $Y$  conditionally by

$$\mathbb{P}_r(Y > y \mid U_1 = u_1, U_2 = u_2) = y^{-1/\gamma_r(u_1, u_2)}, \quad y \geq 1, \quad r \in \{0, 1\}.$$

All stated regularity conditions hold with  $c \equiv 1$ . Coordinate 1 is inactive under  $P_0$  and active under  $P_1$ .

For every  $u_1$  and  $y \geq 1$ ,

$$\begin{aligned} \mathbb{P}_1(Y > y \mid U_1 = u_1) &= \int_0^1 y^{-1/g\{(u_1+v) \bmod 1\}} dv \\ &= \int_0^1 y^{-1/g(v)} dv = \mathbb{P}_0(Y > y \mid U_1 = u_1), \end{aligned}$$

where the middle identity is invariance of Lebesgue measure under rotation of the circle. Hence the joint laws of  $(U_1, Y)$  coincide. The distribution of  $\phi_n$  is therefore the same under  $P_0$  and  $P_1$ ; if its common probability of outputting one is  $q$ , the two errors are  $q$  and  $1 - q$ , whose sum is one.  $\square$

## 10 Group order: the exact price of escaping marginal invisibility

For  $J \subseteq \{1, \dots, p\}$  define, under full support,

$$\xi_J(u_J) = \max_{v \in [0,1]^{p-|J|}} \gamma(u_J, v), \quad \Delta_J = \gamma^* - \int_{[0,1]^{|J|}} \xi_J(u_J) du_J.$$

**Theorem 10.1** (Group detectability and minimal interaction order). *Assume full fibre support and continuity of  $\gamma$ . For every nonempty  $J$ ,*

$$\Delta_J > 0 \iff \lambda_{|J|}([0,1]^{|J|} \setminus \pi_J(\mathcal{M})) > 0. \quad (47)$$

For an active coordinate  $j$ , define

$$r_*(j) = \min\{|J| : j \in J, \Delta_J > 0\}.$$

If  $\mathcal{A}$  is the active set and  $s = |\mathcal{A}|$ , then  $r_*(j) \leq s$  for every  $j \in \mathcal{A}$ . The bound is sharp: for

$$\gamma(u) = g\{(u_1 + \dots + u_s) \bmod 1\},$$

with nonconstant positive periodic continuous  $g$ , one has  $r_*(j) = s$  for every  $j \leq s$ .

*Proof.* The proof of (??) is identical to the one-coordinate argument:  $\xi_J(u_J) = \gamma^*$  exactly when the  $J$ -fibre through  $u_J$  intersects  $\mathcal{M}$ , equivalently when  $u_J \in \pi_J(\mathcal{M})$ ; the integrated nonnegative deficit is positive exactly when the complement has positive measure.

Take  $J = \mathcal{A}$ . Then  $\xi_{\mathcal{A}}(u_{\mathcal{A}}) = \gamma(u_{\mathcal{A}})$  because no active coordinate is maximized out. Since  $j \in \mathcal{A}$ , the continuous function  $\gamma$  is not constant on  $[0,1]^{\mathcal{A}}$ , so the open set on which  $\gamma < \gamma^*$  is nonempty and has positive measure. Thus  $\Delta_{\mathcal{A}} > 0$  and  $r_*(j) \leq s$ .

For the cyclic-sum example, fix any  $J$  with  $|J| < s$  and any value  $u_J$ . At least one active coordinate is omitted. Choose that omitted coordinate, modulo one, so that the total sum is a maximizer of  $g$ ; set the other omitted coordinates arbitrarily. Hence every  $u_J$  belongs to  $\pi_J(\mathcal{M})$ , so  $\Delta_J = 0$ . For  $J = \mathcal{A}$ , the preceding paragraph gives  $\Delta_{\mathcal{A}} > 0$ . Therefore  $r_*(j) = s$ .  $\square$

## 11 Summary of the strengthened theorem chain

The exact population target is  $\mathcal{D}$ , the positive-gap set generated by the conditional supports. Under null argmax accessibility,  $\mathcal{D} \subseteq \mathcal{A}$ ; it equals  $\mathcal{A}$  precisely when every active coordinate misses the accessible argmax on a positive-measure set. Continuity of projected conditional laws is unnecessary. The original empirical-rank score consistently recovers  $\mathcal{D}$  under a direct quadrature condition, with the rank-displacement requirement weakened from  $\log(pn) = o(nh^2)$  to  $\log p = o(nh^2)$ . The disjoint-block score further improves the stochastic score rate from

$$\sqrt{\frac{\log(pn)}{n\alpha h}} \quad \text{to} \quad \sqrt{\frac{\log p}{n\alpha}},$$

while replacing uniform-in-location tail and spatial assumptions by their block averages. Finally, primitive deficit-mass doubling can be replaced by the strictly weaker uniform Laplace-localization condition, and the impossibility theorem shows that no purely coordinatewise marginal method can remove the remaining argmax-projection limitation.